

Documentation

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1 Overview

The primary purpose of this document is to generate the equations needed in the documentation

2 Solar Controller

Controller efficiency equation

$$\eta_{\text{controller}} = \frac{V_{\text{out}} I_{\text{out}}}{V_{\text{in}} I_{\text{in}}}$$

Panel efficiency

$$\eta = \frac{P_{\text{out}}}{E_{\text{light}} A}$$

3 PR Detector

Atmospheric depth

$$X = \int_h^\infty \rho(h') dh'$$

Hydrostatic equilibrium

$$\frac{dP}{dh} = -\rho g$$

Atmospheric depth in terms of pressure

$$X = \frac{P}{g}$$